

什么是 CLIP?

CLIP 的全称是 **Contrastive Language-Image Pre-training** (对比文本-图像预训练)。它是 OpenAI 在 2021 年推出的一种多模态 (Multimodal) 深度学习模型。 PDF + 1

用一句通俗的话来解释它的核心能力: CLIP 建立了一座连接“视觉”和“语言”的桥梁, 让计算机能够像人类一样, 看懂一张图片并用自然语言去理解它, 或者根据一段文字找到对应的图片。 PDF

CLIP 解决了传统计算机视觉的什么痛点?

在 CLIP 出现之前, 传统的图像识别模型 (比如基于 ImageNet 训练的经典 ResNet) 存在两个巨大的局限性:

- 1 **类别固定 (标签受限)**: 传统的模型就像是一个“单选题机器”。如果它在训练时只学过 1000 种卡片 (比如猫、狗、汽车), 那么当你给它看一个新东西 (比如“无人机”), 它就完全抓瞎, 只能强行在已有的 1000 个标签里猜一个最像的。想增加新类别? 对不起, 重新收集数据、打标签、训练模型。 PDF + 1
- 2 **缺乏泛化能力**: 传统的模型高度依赖人工标注的“标准答案”。它们在特定的测试集上表现很好, 但一旦拿到现实世界中, 面对光照、视角或画风的改变 (也就是“分布偏移”), 错误率就会飙升。 PDF

优点: 可扩大 Batch Size
② CLIP 侧重全局语义
SgIP 保留细节


CLIP 的颠覆之处在于: 它直接从互联网上公开的、海量的“图片 + 文字描述” (图文对) 中进行被动学习。它不再绑定于任何固定的类别标签, 而是直接学习“图像”与“人类自然语言”之间的对应关系。

PDF + 1

$N \times N$ 每个都算
Softmax 换成 Sigmoid
= 分类问题

CLIP 是如何工作的? (核心机制)

CLIP 的工作流程非常优雅, 论文中将其分为三个主要步骤: PDF

(1) 对比预训练 (Contrastive Pre-training)

图片 $\rightarrow$ [图像编码器] $\rightarrow$ 图像特征向量 (I) \ 计算余弦相似度
文本 $\rightarrow$ [文本编码器] $\rightarrow$ 文本特征向量 (T) / $\rightarrow$ 寻找对角线上的正确配对

(2) 创建数据集分类器 (Create Dataset Classifier)

类别文字 (“一张[猫]的照片”) $\rightarrow$ [文本编码器] $\rightarrow$ 文本特征向量

(3) 零样本预测 (Zero-shot Prediction)

新图片 $\rightarrow$ [图像编码器] $\rightarrow$ 与所有类别文本权重比对 $\rightarrow$ 输出最高分预测

SgIP

↑ 改

缺陷

1. 对比预训练 (Contrastive Pre-training)

这是 CLIP 最核心的训练方式。 PDF

- o OpenAI 收集了 4 亿对从互联网上抓取的“图片 + 文本描述”数据集 (称为 WIT)。 PDF
- o CLIP 内部包含两个引擎: 一个图像编码器 (Image Encoder) 和一个文本编码器 (Text Encoder)。 PDF
- o **训练任务**: 如果一个 Batch (批次) 里有 N 张图片和 N 个文本, 它们一共可以组合出 $N \times N$ 种可能性。CLIP 的任务极其简单明确——最大化真正原本就配对的 N 个组合的相似度, 同时最小化其他 $N^2 - N$ 个错误组合的相似度。它不需要去逐字预测文本里的单词, 只需要做图文的“连线题”。

需要看以前材料

$\rightarrow$ 双编码器架构

loss 是 InfoNCE loss

```

1 import torch
2 import torch.nn as nn
3 import torch.nn.functional as F
4
5 class SimpleCLIP(nn.Module):
6     def __init__(self, image_encoder, text_encoder, embed_dim, temperature=0.07):
7         super().__init__()
8         self.image_encoder = image_encoder
9         self.text_encoder = text_encoder
10        # 可学习的温度系数，用来控制相似度分布的平滑度
11        self.temperature = nn.Parameter(torch.ones(1) * log(1 / temperature))
12
13    def forward(self, images, text):
14        # 1. 提取特征
15        image_features = self.image_encoder(images) # [N, D]
16        text_features = self.text_encoder(text)     # [N, D]
17
18        # 2. 归一化
19        image_features = F.normalize(image_features, dim=-1)
20        text_features = F.normalize(text_features, dim=-1)
21
22        # 3. 计算余弦相似度并缩放
23        # [N, D] @ [D, N] -> [N, N]
24        logit_scale = self.temperature.exp()
25        logits_per_image = logit_scale * image_features @ text_features.t()
26        logits_per_text = logits_per_image.t()
27
28        # 4. 构建标签（对角线是正确答案，即 0, 1, 2, ..., N-1）
29        labels = torch.arange(images.size(0), device=images.device)
30
31        # 5. 双向交叉熵损失（图像找文本，文本找图像）
32        loss_img = F.cross_entropy(logits_per_image, labels)
33        loss_txt = F.cross_entropy(logits_per_text, labels)
34        loss = (loss_img + loss_txt) / 2
35
36    return loss

```

2. 零样本预测（Zero-shot Prediction）

训练完成后，CLIP 拥有了强大的“盲测”能力，即 **Zero-shot（零样本）迁移能力**。 PDF

- 面对一个全新的图像分类数据集（比如有 10 个类别），CLIP **不需要针对该数据集进行任何二次训练或微调**。 PDF
- **怎么做分类呢？** 我们把这 10 个类别的名称（如：猫、狗、鸟.....）放入一个提示模板中，变成句子（如：“一张猫的照片”、“一张狗的照片”）。
- 文本编码器会将这些句子转化为向量，作为分类器的权重。当有一张新图片输入图像编码器后，模型会计算该图片向量与这 10 个文本向量的相似度，**得分最高的那个文本，就是模型的预测答案**。

为什么 CLIP 这么牛？（主要优势）

- **“免费”的 Zero-shot 能力：** 不需要任何下游任务的标注数据，直接拉上场就能比赛。论文中，零样本的 CLIP 在 ImageNet 数据集上的准确率，直接打平了精心训练的经典 Supervised ResNet-50。 PDF + 1
- **极强的鲁棒性（抗噪/抗魔改）：** 因为 CLIP 是在互联网大熔炉里炼出来的，它学的是语义本身。传统的 ImageNet 模型在面对素描图、对抗样本、或电影截图时（比如一张香蕉的素描），识别率会暴跌。而 CLIP 表现得非常稳定，因为它知道无论怎么变，那都是“香蕉”的语义（有效鲁棒性提升了多达 75%）。 PDF
- **提示词工程（Prompt Engineering）的威力：** 通过改变输入的文本描述，你可以任意微调模型的关注点。比如加入“一张小小的{标签}”或“一幅{标签}的卫星照片”，就能让模型在特定垂直领域的分类准确率大幅提升。

CLIP 有什么局限性吗？

虽然 CLIP 很强，但论文中也坦诚地指出了它的软肋：

- 1 **复杂/抽象任务抓瞎：** 在面对非常精细的分类（如同款车型的不同年份、不同特征的飞机微小型号）或者抽象任务（如同一个画面里数一数有几个苹果）时，CLIP 的零样本表现往往很差，甚至接近随机猜测。
- 2 **对手写数字（MNIST）等极度出圈的数据泛化差：** 有意思的是，CLIP 在复杂的语义上表现惊艳，但让它识别 MNIST 手写数字时，准确率只有 88%，甚至不如最简单的像素回归分类器。原因在于它训练的 4 亿张互联网图片里，几乎没有长得像 MNIST 那种纯粹、干净的手写数字图。
- 3 **社会偏见（Bias）问题：** 因为训练数据直接来源于 unfiltered（未过滤）的互联网图文，CLIP 不可避免地吸收了人类互联网上的性别、种族以及社会刻板印象和偏见（例如在较低置信度下，倾向于将某些特定族群与犯罪、动物等贬损词汇联系在一起）。

总结：用在什么地方？

今天，CLIP 已经成为了多模态 AI 领域的**基础设施**。如果你熟悉以下技术，你会发现到处都有 CLIP 的影子：

- **Stable Diffusion / Midjourney 等文生图大模型：** 它们之所以能听懂你的 Prompt（提示词）并画出对应的图像，正是因为底层有类似 CLIP 的文本编码器在把关，指导画面向正确的语义靠拢。
- **智能相册搜索：** 在手机相册里输入“去年在海边吃冰激凌的照片”，不需要人工打标签，算法就能帮你精准揪出照片，其背后依靠的也是 CLIP 这种图文互检索的能力。

简而言之，CLIP 让视觉模型拥有了人类语言的常识，是计算机视觉迈向通用人工智能（AGI）的一大步。 PDF

Learning Transferable Visual Models From Natural Language Supervision

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Abstract

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study the performance of this approach by benchmarking on over 30 different existing computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we match the accuracy of the original ResNet-50 on ImageNet zero-shot without needing to use any of the 1.28 million training examples it was trained on. We release our code and pre-trained model weights at <https://github.com/OpenAI/CLIP>.

1. Introduction and Motivating Work

Pre-training methods which learn directly from raw text have revolutionized NLP over the last few years (Dai & Le, 2015; Peters et al., 2018; Howard & Ruder, 2018; Radford et al., 2018; Devlin et al., 2018; Raffel et al., 2019).

Task-agnostic objectives such as autoregressive and masked language modeling have scaled across many orders of magnitude in compute, model capacity, and data, steadily improving capabilities. The development of “text-to-text” as a standardized input-output interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets removing the need for specialized output heads or dataset specific customization. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009). Could scalable pre-training methods which learn directly from web text result in a similar breakthrough in computer vision? Prior work is encouraging.

Over 20 years ago Mori et al. (1999) explored improving content based image retrieval by training a model to predict the nouns and adjectives in text documents paired with images. Quattoni et al. (2007) demonstrated it was possible to learn more data efficient image representations via manifold learning in the weight space of classifiers trained to predict words in captions associated with images. Srivastava & Salakhutdinov (2012) explored deep representation learning by training multimodal Deep Boltzmann Machines on top of low-level image and text tag features. Joulin et al. (2016) modernized this line of work and demonstrated that CNNs trained to predict words in image captions learn useful image representations. They converted the title, description, and hashtag metadata of images in the YFCC100M dataset (Thomee et al., 2016) into a bag-of-words multi-label classification task and showed that pre-training AlexNet (Krizhevsky et al., 2012) to predict these labels learned representations which preformed similarly to ImageNet-based pre-training on transfer tasks. Li et al. (2017) then extended this approach to predicting phrase n-grams in addition to individual words and demonstrated the ability of their system to zero-shot transfer to other image

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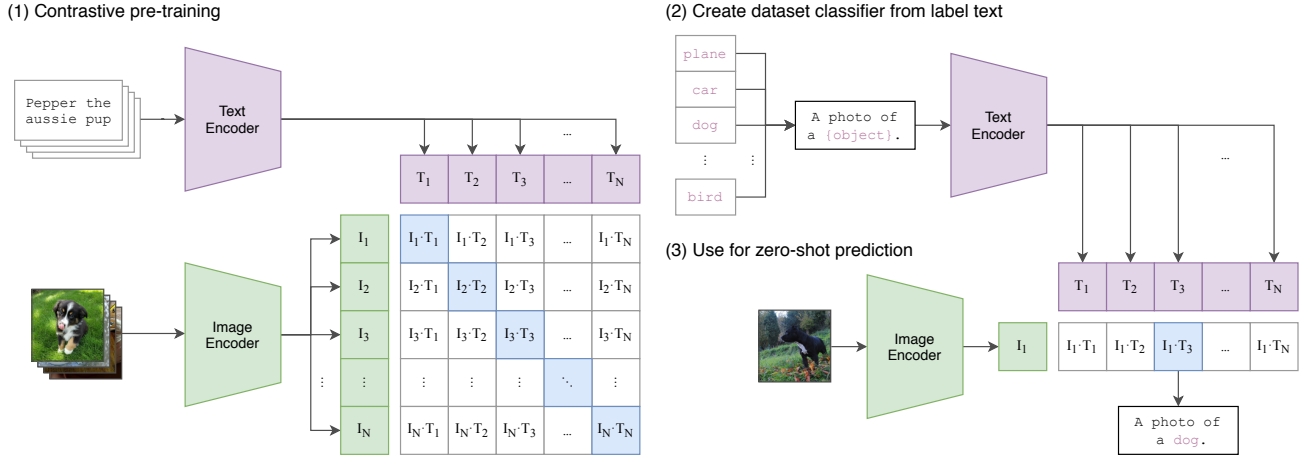


Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset’s classes.

classification datasets by scoring target classes based on their dictionary of learned visual n-grams and predicting the one with the highest score. Adopting more recent architectures and pre-training approaches, VirTex (Desai & Johnson, 2020), ICMLM (Bulent Sariyildiz et al., 2020), and ConVIRT (Zhang et al., 2020) have recently demonstrated the potential of transformer-based language modeling, masked language modeling, and contrastive objectives to learn image representations from text.

While exciting as proofs of concept, using natural language supervision for image representation learning is still rare. This is likely because demonstrated performance on common benchmarks is much lower than alternative approaches. For example, Li et al. (2017) reach only 11.5% accuracy on ImageNet in a zero-shot setting. This is well below the 88.4% accuracy of the current state of the art (Xie et al., 2020). It is even below the 50% accuracy of classic computer vision approaches (Deng et al., 2012). Instead, more narrowly scoped but well-targeted uses of weak supervision have improved performance. Mahajan et al. (2018) showed that predicting ImageNet-related hashtags on Instagram images is an effective pre-training task. When fine-tuned to ImageNet these pre-trained models increased accuracy by over 5% and improved the overall state of the art at the time. Kolesnikov et al. (2019) and Dosovitskiy et al. (2020) have also demonstrated large gains on a broader set of transfer benchmarks by pre-training models to predict the classes of the noisily labeled JFT-300M dataset.

This line of work represents the current pragmatic middle ground between learning from a limited amount of supervised “gold-labels” and learning from practically unlimited amounts of raw text. However, it is not without compro-

mises. Both works carefully design, and in the process limit, their supervision to 1000 and 18291 classes respectively. Natural language is able to express, and therefore supervise, a much wider set of visual concepts through its generality. Both approaches also use static softmax classifiers to perform prediction and lack a mechanism for dynamic outputs. This severely curtails their flexibility and limits their “zero-shot” capabilities.

A crucial difference between these weakly supervised models and recent explorations of learning image representations directly from natural language is scale. While Mahajan et al. (2018) and Kolesnikov et al. (2019) trained their models for accelerator years on millions to billions of images, VirTex, ICMLM, and ConVIRT trained for accelerator days on one to two hundred thousand images. In this work, we close this gap and study the behaviors of image classifiers trained with natural language supervision at large scale. Enabled by the large amounts of publicly available data of this form on the internet, we create a new dataset of 400 million (image, text) pairs and demonstrate that a simplified version of ConVIRT trained from scratch, which we call CLIP, for Contrastive Language-Image Pre-training, is an efficient method of learning from natural language supervision. We study the scalability of CLIP by training a series of eight models spanning almost 2 orders of magnitude of compute and observe that transfer performance is a smoothly predictable function of compute (Hestness et al., 2017; Kaplan et al., 2020). We find that CLIP, similar to the GPT family, learns to perform a wide set of tasks during pre-training including OCR, geo-localization, action recognition, and many others. We measure this by benchmarking the zero-shot transfer performance of CLIP on over 30 existing datasets and find